VERSION
14.0

AMS GRA ARCHITECTURE VOLUMES

Time Distribution

AFLCMC/WA

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1.0 Overview

Time distribution enables mission capabilities by synchronizing the various elements within a Mission System and ensuring all of the subsystems are operating on a common clock.

1.1 Purpose

The Time Distribution Volume describes the time distribution options a Mission System Integrator (MSI) is allowed to implement within an Agile Mission Suite (AMS) Government Reference Architecture (GRA)-compliant Mission System.

1.2 Scope

The Time Distribution Volume aids MSIs in designing an AMS GRA-compliant time distribution system. This time volume can also be used by any entity seeking an understanding of the time distribution options that are used by Minimum Procurable Units (MPUs) in an AMS GRA-compliant Mission System.

1.3 Informative References

Informative references are supplemental documents that a procuring authority, MSI, or MPU provider refers to for more information about an AMS GRA entity or related entities. See **Table 1.3.1** for a list of informative references for the Time Distribution Volume.

Standard	Title	Version/Date
AMS GRA Volumes	Mission System Volume	14.0
IEEE 1588-2008	IEEE Standard for a Precision Clock Synchronization Protocol for Networked Measurement and Control Systems	2008-07-24
OMS Standard	OMS Description and Documentation	

Table 1.3.1: Informative References Identify Documents a Procuring Authority, MSI, or MPU Provider Can Use for More Information About Time Distribution and Related Entities.

2.0 Time

AMS GRA supports a distributed, heterogeneous network of apertures, sensors, and compute resources. In order to optimally manage these resources, the underlying clocks of the sensors and compute resources need to share a common time base. As such, it is critical for AMS GRA to support some form of ownship time synchronization. This synchronization capability enables scheduling of the distributed resources in a manner that avoids temporal conflicts that would arise in an unsynchronized system. Additionally, synchronization of sensor observables is a mission capability enabler, with an increasing variety of capabilities available as the precision of synchronization increases. As such, AMS GRA supports a number of time synchronization interfaces, in particular Precision Time Protocol (PTP), Distributed Guaranteed Time (DGT), and Open Mission Systems (OMS) Legacy Timing. Additional details on these protocols are shown in **Figure 2.0.1**. AMS GRA does not impose an architectural requirement on the accuracy of time synchronization achieved by the mixture of supported precision time interfaces, as this is subject to the requirements of specific Mission Systems.

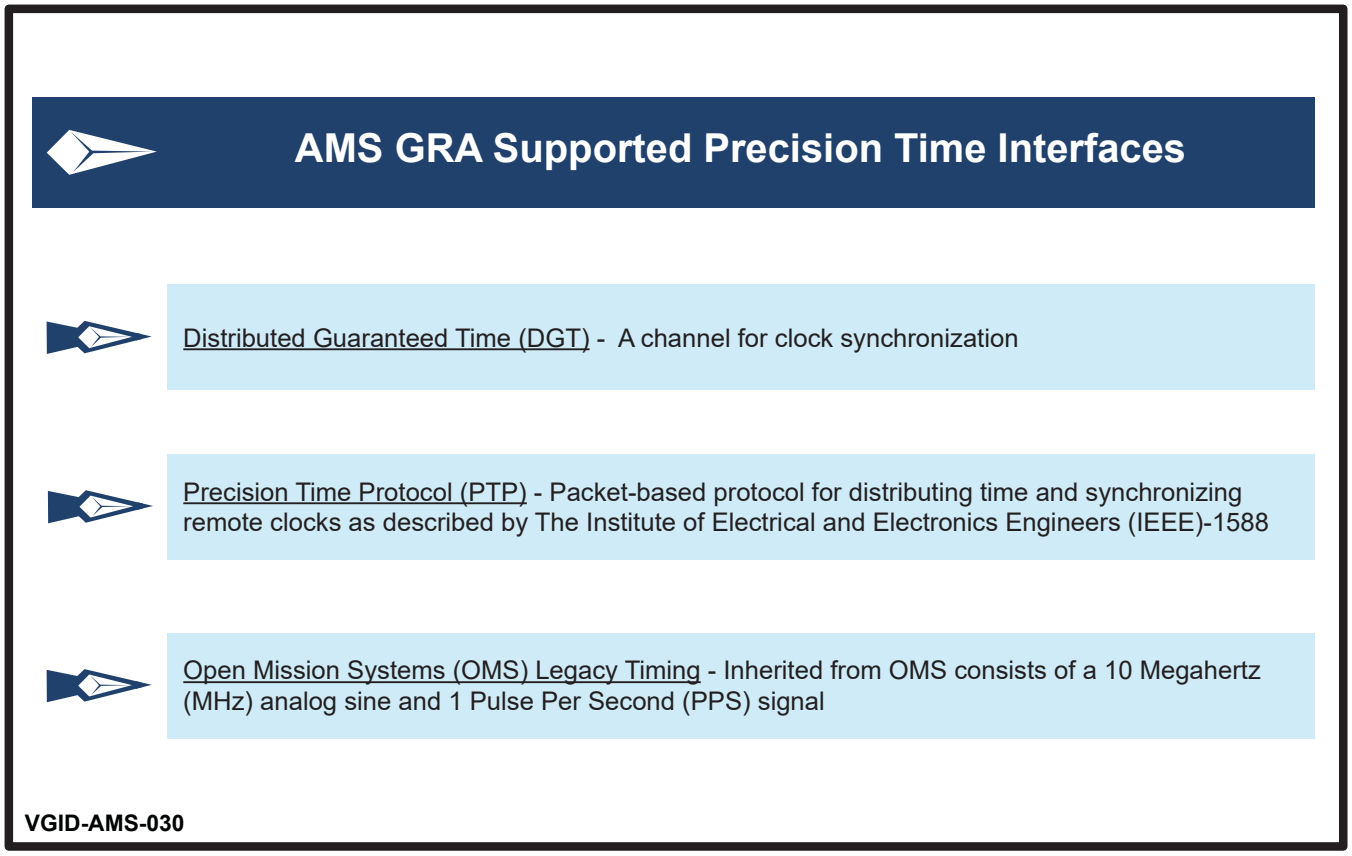


Figure 2.0.1: Time Synchronization Across Distributed Resources is Critical to Prevent Scheduling Conflicts within the Platform.

The Mission Platform is responsible for maintaining time. The AMS GRA Level 2 and Level 3 MPUs will receive time using one or more mechanisms defined in **Figure 2.0.1**, in alignment with platform time distribution implementation. Software Service MPUs obtain time from the Digital Payload Processor (DPP) Application Programming Interface (API).

The MSI is responsible for both the implementation and integration of multiple time distribution mechanisms. One approach may leverage the use of Global Positioning System (GPS) to synchronize an on-board clock. The system time source receives the GPS 1 Pulse Per Second (PPS) (and optional GPS 10 Megahertz (MHz)) from the Aircraft Navigation System which is assumed to include a GPS receiver. The time source drives the Legacy 1 PPS and Legacy 10 MHz outputs, DGT, and/or PTP, all from the same internal reference. All of these distribution mechanisms have the same nominal time, but different capabilities in terms of offset error or jitter to this reference.

Synchronization of the three precision timing interfaces is left to the specific implementation of the Mission Platform architecture.

Figure 2.0.2 shows a notional example of how the 3 applicable time distribution methods may distribute time from the platform to AMS GRA MPUs.

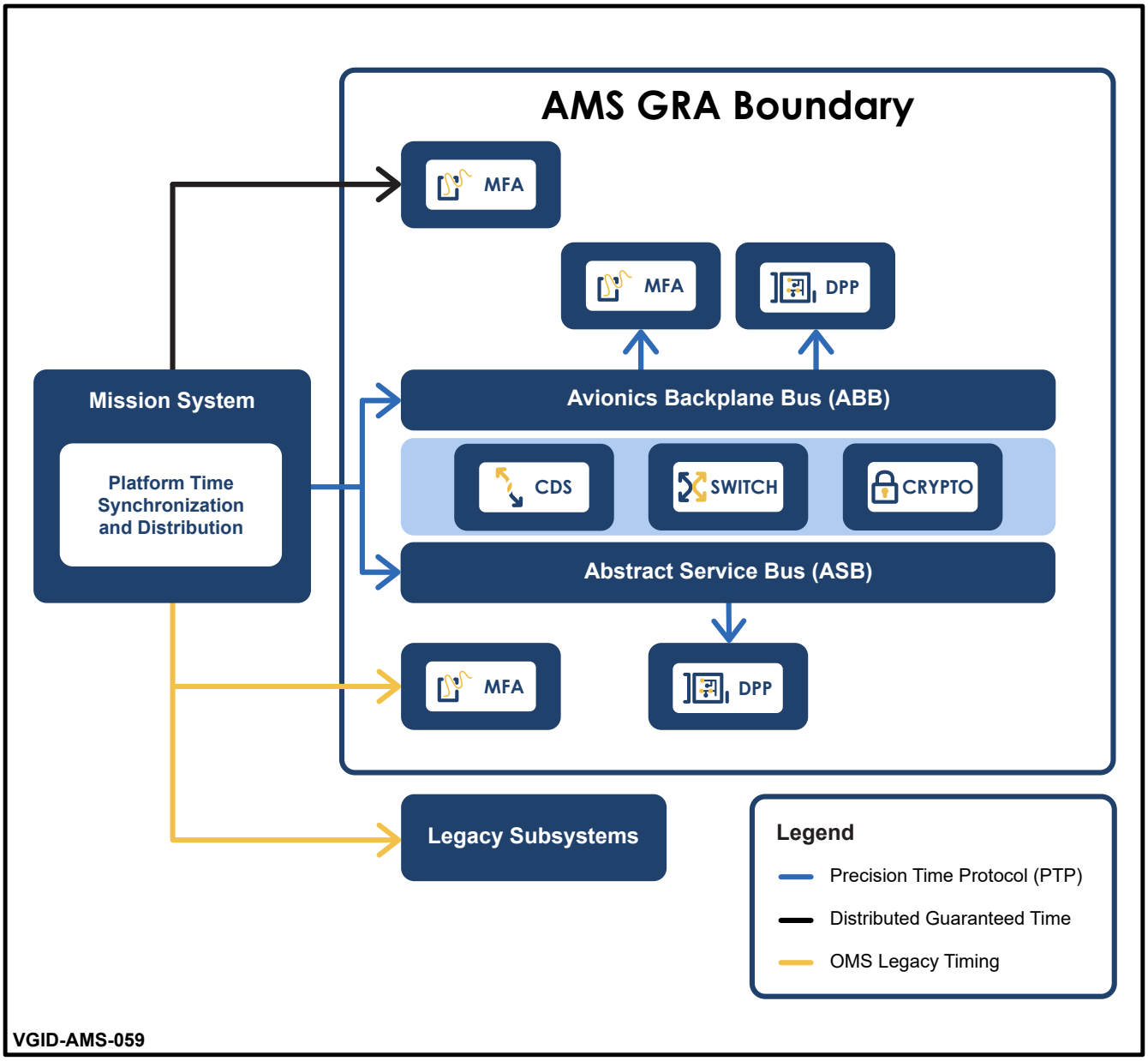


Figure 2.0.2: The AMS GRA Supports Three Possible Time Distribution Methods. Different Methods Are Used for Different Purposes and Allow for Different Levels of Time Accuracy.

DGT coordinates time between the platform time source and Radio Frequency (RF) Multi-Function Apertures (MFAs). For additional details, refer to the DGT Interface Description Document (IDD).

PTP is provided over the Avionics Backplane Bus (ABB) and Abstract Service Bus (ASB) network sourced by the system time source (server). PTP can have accuracy in nanoseconds to microseconds depending on use of hardware or software-based time stamping. System behavior operates on a pipeline-processing approach so that:

- What is planned to transmit in the future is being scheduled and described in In-phase/Quadrature (I/Q) data and MFA setup now
- What was planned in the past is transmitting/receiving now
- What was transmitted/received in the past is being processed now

The pipeline-processing approach allows lower precision throughout all signal processing elements. For example, the RF MFA needs to have the high precision for the tasks of time stamping I/Q data upon receiving or transmitting at the exact time the I/Q data is marked. As long as the DGT and PTP time systems operate at the same frequency, the timeline does not exhibit slipping. Since both PTP and DGT are sourced from the same reference, they are forced to the same frequency.

Finally, the OMS Legacy 1 PPS and 10 MHz that are clients to the time source can be also provided for MFAs or legacy systems that use this interface and require synchronization with the overall time system. These MFAs are limited in accuracy at the nanoseconds level depending on the degree to which the system can calibrate the one-way 1 PPS delay.

2.1 Distributed Guaranteed Time (DGT)

The DGT interface is an AMS GRA defined method of distributing mission system time.

2.2 Precision Time Protocol (PTP) Network

PTP is a messaging protocol designed to work over packet-based networks. The PTP standard does not rely on physical layer network standards such as Ethernet. PTP functions over both Ethernet and Infiniband.

Figure 2.2.1 shows a notional example of how the PTP ports might logically map to the relevant MPUs in AMS GRA. Neither the figure nor the descriptions of supported time interfaces prescribe which time interfaces are required to be used to support time distribution. This functionality allows flexibility to adopting programs. A timing source serves as the default PTP network server. The platform Position Navigation and Time (PNT) physically connects to the Switch MPU to distribute PTP on the ASB and ABB. The Switch MPUs must be PTP aware and

implement either a PTP Transparent clock or PTP Boundary clock. Ordinary PTP clocks reside within the MFAs and DPPs. These ordinary clocks nominally serve as client clocks, but in the event of a failure of the default PTP server, an ordinary clock can serve as a secondary PTP server.

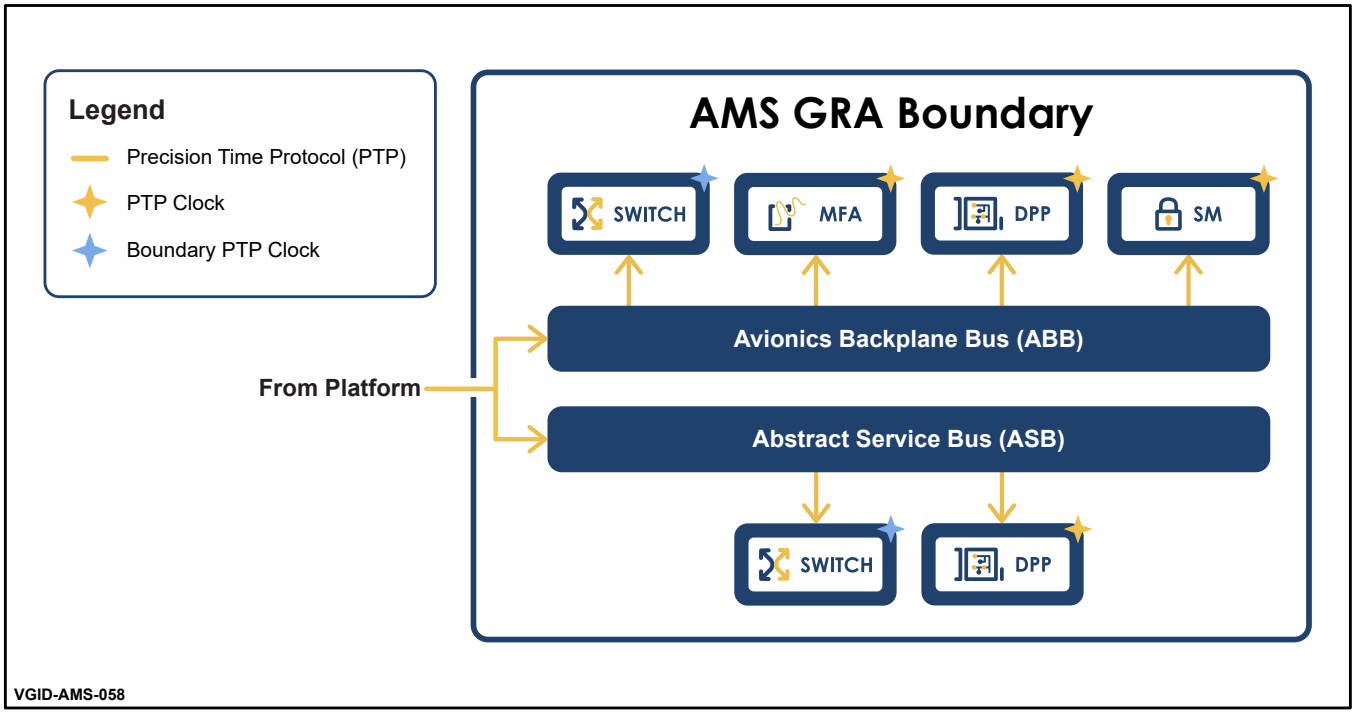


Figure 2.2.1: The Packet-Based Protocol Design of PTP Allows the ABB and ASB Network to Provide Flexibility to Adopting Programs in Getting Time Distributed to Applicable MPUs.

Not all MPUs are required to have a PTP port. For example, some RF MFAs gain no benefit with the timing accuracy and precision provided by PTP. In this case, precise time is obtained by the DGT or OMS legacy interface. On the other hand, all AMS compliant DPPs have a PTP port. See requirements GRA-MS-004, GRA-MPU-014, and GRA-MPU-015.

For MPUs with a PTP port, a distinction is made between PTP time that is kept by the PTP Network Interface Card (NIC) and local MPU system clock time (see **Figure 2.2.1**). The PTP function running on an MPU is allowed to synchronize the MPU system clock to PTP time.

2.3 OMS Legacy Timing

The Legacy OMS Timing Interface is defined in the OMS Definition and Documentation Volume. This interface consists of a time message and a discrete signal. The time message is read at the rising edge of the discrete signal, which is the moment when the discrete signal transitions from its low to high state. This discrete signal pulses on and off at a rate of 1 PPS given a 10 MHz clock reference. The 10 MHz clock reference is a 1 volt peak sinusoid into a nominal 50 ohm characteristic impedance. The 10 MHz signal is phase coherent to the associated 1 PPS signal.

The OMS Legacy timing can be traced back through multiple systems. The AMS GRA system receives the legacy timing information from the Aircraft's Navigation System. The Aircraft's Navigation System is assumed to receive its information from the Aircraft System's time source.

The Aircraft's Navigation system uses the 1 PPS discrete signal as a minimum requirement to maintain knowledge of GPS time when available. Additionally, the use of the 10 MHz reference signal is optional based on the implementation within a given time reference.

The OMS time messages and discrete signal are connected to the relevant MPUs. These two inputs are then optionally used by MFAs that have this legacy time interface for time synchronization. If an MFA requires phase or time coherence, the MFA itself provides a calibration mechanism to maintain the required coherence.

3.0 Compliance

The specific time requirements for AMS GRA are held in the Mission System and MPU Volumes. The MSI is responsible for supporting any of the time interfaces specified above as required by the MPUs. Switch and DPP MPUs are required to support PTP at various levels based on the tier of compliance. See the associated volumes for compliance details.

A.0 Acronyms

ABB	Avionics Backplane Bus
AMS	Agile Mission Suite
API	Application Programming Interface
ASB	Abstract Service Bus
DGT	Distributed Guaranteed Time
DPP	Digital Payload Processor
GPS	Global Positioning System
GRA	Government Reference Architecture
I/Q	In-phase/Quadrature
IDD	Interface Description Document
IEEE	The Institute of Electrical and Electronics Engineers
MFA	Multi-Function Aperture
MHz	Megahertz
MPU	Minimum Procurable Unit
MSI	Mission System Integrator
NIC	Network Interface Card
OMS	Open Mission Systems
PNT	Position Navigation and Time
PPS	Pulse Per Second
PTP	Precision Time Protocol
RF	Radio Frequency
SK	Starter Kit

B.0 Revision History

The Revision History table, **Table B.0.1**, captures the major changes made for a given AMS GRA volume as the Starter Kit (SK) continues to mature.

Revision	Description	Date	Notes
-	This revision adds a revision table to the volume.	May 22, 2024	SK11.0
A	This revision did not introduce any significant changes.	July 24, 2025	SK13.0
B	This revision did not introduce any significant changes.	April 24, 2026	AMS GRA 14.0

Table B.0.1: Revision History Captures the Major Changes Made to the Time Distribution Volume Between Documentation Releases.